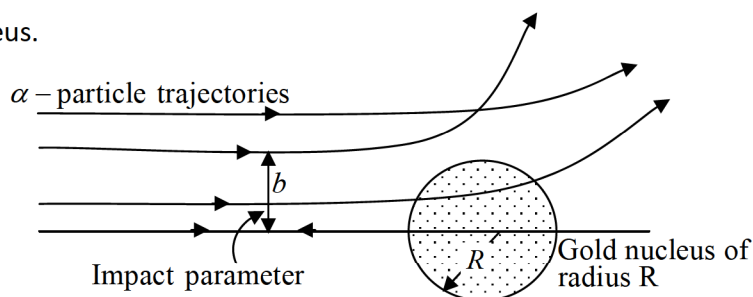


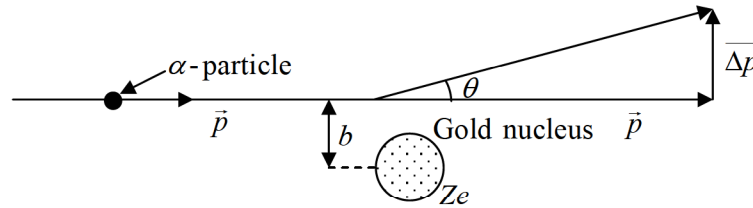
Chapter One

Basic Nuclear Properties

1.1 Rutherford Experiment and Properties of Nucleus

You are all familiar with the structure of an atom. The modern picture of an atom tells us that there is a core called the nucleus, which contains all the positive charge, and almost all the mass of the atom. This nucleus has positively charged particles called the protons in it, and electrically neutral neutrons. In the first part of this course it is this core called the nucleus that will occupy our time. In an atom, which is electrically neutral there is equal amount of negative charge in it. The negatively charged electrons are all distributed in different atomic orbitals around this nucleus.





Principle of Rutherford experiment:

By bombarding a very thin gold foil with α -particles, there were an observation that a small fraction of these α -particles were deflected at large angle.

This angle is defined b/w the asymptotic direction of incident particle and asymptotic line of deported α -particle which is known as scattering angle (θ).

And the minimum distance at which α -particle would approach the nucleus if there were no force between them is known as impact parameter (b)

And on this impact parameter and radius of the nucleus there is dependence of change in deflection, which is categorized as:-

- For $b < r$ (Ray 2), the deflection produced in the α -particle trajectory is small.
- For $b > r$ (Ray 4) due to less coulomb repulsion the deflection of α -particle is also less.
- And for $b \sim r$ (Ray 3) the deflection of α -particle becomes maximum.
- There is also a deflection at one point (Ray 1) where the KE of α -particle will start converting into the coulombian potential energy and at one point its KE will become 0 and that point will come back due to repulsive force and the distance b/w that point and nuclei is known as distance of closest approach.

Conclusion:

- Most of the space inside the atom is empty.
- As a few α -particles got deviated through small angles there must be some positively charged particles in matter which repel the α -particles strongly causing deviation and there positively charged particles known as nucleus.
- As a very small number of α -particles got deviated through large angles and retraced their path, the size of the nucleus must be very small compared to that of an "atom".
The nucleus must also be very heavy.

Calculation of the repulsive force

$$F = \frac{e^2}{4\pi\epsilon_0} \cdot \frac{2z}{b^2}, \quad \frac{e^2}{4\pi\epsilon_0} = 1.44 \text{ MeV-fm}$$

Calculation of distance of closest approach (x)

$$K = \frac{e^2}{4\pi\epsilon_0} \cdot \frac{2z}{x}$$

Calculation of impact parameter:

$$b = \frac{ze^2}{4\pi\epsilon_0 k} \cot \frac{\theta}{2}$$

where k is the KE of α -particle and θ is scattering angle.

Angular scattering distribution.

$$N(\theta) = \frac{N_i n L z^2 k^2 e^4}{4r^2 KE^2 \sin^4 \frac{\theta}{2}}$$

where N_i = number of incident α -particles

n = atoms per unit volume in target

L = thickness of target

z = atomic number of target

e = electronic charge

k = coulomb's constant

r = target to deflector distance

KE = KE of α -particle

θ = scattering angle.

Experiments on scattering of α particle on gold foil shows there is existence of nucleus at the centre of atom. The experiments demonstrate radius of a nucleus is smaller than the radius of atom by factor of about 10^4 . This means the volume of an atom is about 10–12 times the volume of the nucleus. In other words, an atom is almost empty.

Atomic mass: The mass of atom is very small compare to kilogram .for example mass of carbon atom ^{12}C is $1.99 \times 10^{-26} \text{ kg}$.Kilogram is not a very convenient unit to measure such small quantities .therefore a different mass unit is used for expressing atomic masses. This unit is the atomic mass unit (u), defined as $\frac{1}{12}$ of the mass of the carbon ^{12}C

$$1u = \frac{\text{mass of one } ^{12}\text{C atom}}{12} = \frac{1.99 \times 10^{-26}}{12} = 1.66 \times 10^{-27} \text{ kg}$$

Example: Mass of iron is $55.85u$. What is equivalent mass in kg?

Solution: Mass of iron is $55.85u = 55.85 \times 1.66 \times 10^{-27} \text{ kg} = 9.26 \times 10^{-26} \text{ kg}$

Composition of atom and nucleus: classical model of atoms suggest **any** atom consist of electron and nucleus. Electron is negatively charged particle with mass $m_e = 0.00055u$. According to classical theory electron is orbiting about nucleus under influence of electrostatic force. The nucleus consists of Proton and neutron. Proton is discovered by Earnest Rutherford. Proton is positively charged same magnitude of electron and it has mass of $m_p = 1.0072u = 1.67 \times 10^{-27} \text{ kg}$ which is 1836-time mass of an electron. The hydrogen atom has only proton as its nucleus. Although over 99.9 percent of hydrogen nuclei are just single protons, a few also contain another particle named as neutron, and a very few two neutrons, along with the protons. The neutron is discovered by Chadwick. Neutron is neutral particle and it has mass $m_n = 1.0086u = 1.6749 \times 10^{-27} \text{ kg}$ which is almost same as mass of proton.

Other than hydrogen atom all other atoms has proton and neutron in its nucleus.

The conventional symbols for nuclear species, or nuclides, follow the pattern ${}_Z^AX$, where X = chemical symbol of the element.

Z = atomic number of the element which is equal to number of protons in the nucleus

A = mass number of the nuclide number of nucleons in the nucleus.

That means mass number is equal to number of protons plus number of neutrons.

Number of neutron $N = A - Z$

Mass Defect: Mass defect is the difference between the actual atomic mass and the predicted mass calculated by adding the mass of protons and neutrons present in the nucleus. The actual atomic mass is less than the predicted mass calculated by adding the masses of nucleons. This additional mass is accounted for by binding energy that is released when a nucleus is formed. When a nucleus is formed, some of the mass is converted to energy and this result in the mass defect. Due to this reason, the actual mass of an atomic nucleus is less than the mass of particles it is made up of.

The mass defect is given by $\Delta M = (Zm_p + Nm_n) - M_A$ where Z is number of protons N is number of neutrons and M_A actual mass of nucleus.

Mass Energy Equivalence: Einstein showed from his theory of special relativity that it is necessary to treat mass as another form of energy Einstein gave the famous mass-energy

equivalence relation $E = mc^2$ where m is relativistic mass of particle and c is speed of light which is given by $3 \times 10^8 \text{ m/sec}$. If rest mass is given by m_0 . The relativistic mass is given by

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

the total energy is given by $E = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$

Relativistic momentum is given by Where $p = mv = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$

In term of momentum p the total relativistic energy is given by $E = \sqrt{p^2 c^2 + m_0^2 c^4}$.

1gm of substance is equivalent to $1 \times 10^{-3} \text{ kg} \times (3 \times 10^8)^2 = 9 \times 10^{13} \text{ J}$ which means if 1g of matter is converted into energy, there is release of $9 \times 10^{13} \text{ J}$ of energy.

1 Joule is equivalent to $6.24 \times 10^{18} \text{ eV} = 6.24 \times 10^{12} \text{ MeV}$. So 1g of mass is equivalent to

$$9 \times 10^{13} \text{ J} \times 6.24 \times 10^{18} = 56 \times 10^{31} \text{ eV}$$

1 atomic mass unit (1u) is equivalent to $\frac{931 \text{ MeV}}{c^2}$

Example: A Gold nucleus has chemical composition ${}^{197}_{79}\text{Au}$.

- Find the atomic number, mass number, number of proton and number of neutron.
- Find the predicted mass of ${}^{197}_{79}\text{Au}$ nucleus in atomic mass unit. Find the predicted mass of ${}^{197}_{79}\text{Au}$ in kg.
- If actual mass of ${}^{197}_{79}\text{Au}$ is $196.9665u$, find value of mass defect in kg.
- Find the value of mass defect in unit of $\frac{\text{MeV}}{c^2}$.
- Find the value of mass defect in MeV.

Solution: (a) atomic number $Z = 79$

Mass number $A = 197$

Number of protons = atomic number $Z = 79$

Number of neutrons $N = A - Z = 197 - 79 = 118$

(b) The mass of nucleus is

$$79 \times m_p + 118 \times m_n = 79 \times 1.00727u + 118 \times 1.00866u = 198.5962u$$

Mass of nucleus of ${}^{197}_{79}\text{Au}$ in kg $198 \times 1.67 \times 10^{-27} = 331.6 \times 10^{-27} \text{ kg} = 33.16 \times 10^{-26} \text{ kg}$

(c) Predicted mass $198.5962u$ actual mass is $196.9665u$

Mass defect is $\Delta M = 198.596u - 196.966u = 1.624u$

$$= 1.62 \times 1.67 \times 10^{-27} \text{ kg} = 2.62 \times 10^{-27} \text{ kg} = 0.2 \times 10^{-26} \text{ kg}$$

(d) If mass defect is $1.624u$ which is equivalent to $\frac{1.62 \times 931.5}{c^2} = \frac{1509 \text{ MeV}}{c^2}$

(e) equivalent energy for mass defect $1.624u \times c^2 = \frac{1509 \text{ MeV}}{c^2} \times c^2 = 1509 \text{ MeV}$

Some rest masses in various units are:

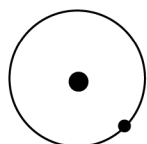
Particle	Mass(kg)	Mass(u)	Mass(MeV/c ²)
Proton	1.6726×10^{-27}	1.007276	938.28
Neutron	1.6750×10^{-27}	1.008665	939.57
Electron	9.1095×10^{-31}	5.486×10^{-4}	0.511
${}_1\text{H}^1$	1.6736×10^{-27}	1.007825	938.79

Nuclear Terminology

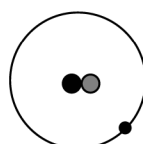
- Isotopes:** If two nuclei have same atomic number Z (proton), then they are called as isotopes. Example: ${}^{13}_6\text{C}$ & ${}^{14}_6\text{C}$, ${}^{16}_8\text{O}$ & ${}^{17}_8\text{O}$ and ${}^1_1\text{H}$, ${}^2_1\text{H}$, ${}^3_1\text{H}$

The varieties of an element that differ in the numbers of neutrons their nuclei contain are called **isotopes**.

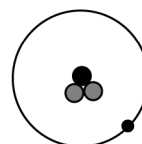
Ordinary hydrogen



Deuterium



Tritium



Proton



Neutron



Electron

Figure: The isotope of hydrogen

- Isotones:** If two nuclei have same neutron number N (proton), then they are called as isotones. Example: ${}^{13}_6\text{C}$ and ${}^{14}_7\text{N}$
- Isobars:** If two nuclei have same mass number A , then they are called as isobars. Example: ${}^{14}_6\text{C}$ and ${}^{14}_7\text{N}$
- Mirror nuclei:** Nuclei with same mass number A but with proton and neutron number interchanged and their difference is ± 1 .

Example: ${}^{11}_6\text{C}$ & ${}^{11}_5\text{B}$ and ${}^{13}_7\text{N}$ & ${}^{13}_6\text{C}$

Size of the nucleus

Radius of Nucleus: By performing scattering experiments in which fast electrons, instead of α -particles, are projectiles that bombard targets made up of various elements, the sizes of nuclei of various elements have been accurately measured. It has been found that a nucleus of mass number A has a radius $R = R_0 A^{1/3}$ where $R_0 = 1.2 \times 10^{-15} \text{ m} = 1.2 \text{ fermi}$.

Volume of Nucleus: This means the volume of the nucleus, which is proportional to R^3 is proportional to A .

Mass Density of Nucleus: Thus the density of nucleus is a constant, independent of A , for all nuclei. Different nuclei are like drop of liquid of constant density.

Assuming spherical symmetry, volume of nucleus is given by $V = \frac{4}{3} \pi R^3 = \frac{4}{3} \pi R_0^3 A$.

Mass of one proton = $1.67 \times 10^{-27} \text{ kg}$, Nuclear Mass = $A \times 1.67 \times 10^{-27} \text{ kg}$.

$$\text{Nuclear density} = \frac{A \times 1.67 \times 10^{-27}}{\frac{4}{3} \pi R_0^3 \times A} = 2.3 \times 10^{17} \text{ kg/m}^3$$

Nucleon Particle Density

Nuclear particle density is ratio of nuclear mass density to nuclear mass per nucleon.

Nuclear mass density is approximately 10^{17} kg/m^3 and nuclear mass per nucleons is $1.67 \times 10^{-27} \text{ kg}$ so

$$\begin{aligned} \text{Nuclear Particle Density} &= \frac{\text{Nuclear Mass Density}}{\text{Nuclear Mass/nucleon}} \\ &= \frac{10^{17} \text{ Kg/m}^3}{1.67 \times 10^{-27} \text{ Kg/Nucleon}} \approx 10^{44} \text{ Nucleons/m}^3 \end{aligned}$$

Example: Given the mass of iron nucleus ${}^{56}_{26}\text{Fe}$ is $55.85u$

- Find radius of nucleus.
- Find volume of nucleus.
- Find mass density of nucleus.
- Find nucleons particle density.

Solution:

(a) Radius is given by $R = R_0 \times A^{1/3}$ where $R_0 = 1.2 \times 10^{-15} \text{ m}$ and $A = 56$

$$1.2 \times 10^{-15} \times (56)^{1/3} = 1.2 \times 10^{-15} \times 3.82 = 4.59 \times 10^{-15} \text{ m}$$

(b) volume of nucleus is $\frac{4\pi}{3} R^3 = \frac{4\pi}{3} \times (R_0)^3 (A^{1/3})^3$

$$= \frac{4\pi}{3} R_0^3 A = \frac{4 \times 3.14}{3} \times (1.2 \times 10^{-15})^3 \times 56 = 405 \times 10^{-45} m^3$$

(c) density is given by $\rho = \frac{m}{V}$

Mass of iron nucleus is $55.85u$ where $u = 1.66 \times 10^{-27} kg$ the mass of iron nucleus is

$$55.8 \times 1.66 \times 10^{-27} kg = 9.27 \times 10^{-26} kg$$

Volume is given by $405 \times 10^{-45} m^3$

Density is $\rho = \frac{m}{V} = \frac{9.27 \times 10^{-26}}{405 \times 10^{-45}} = 0.0228 \times 10^{19} = 2.29 \times 10^{17} kg / m^3$

(d) $n = \frac{\text{Nuclear Mass Density}}{\text{Nuclear Mass/nucleon}} = \frac{2.29 \times 10^{17} kg / m^3}{1.67 \times 10^{-27} kg / nucleon} = 1.37 \times 10^{44} \text{ Nucleon} / m^3$

Example: Electrons of kinetic energy $K = 500 MeV$ are scattered from a target of nuclei of charge distribution radius R into a diffraction pattern that has minima of average separation of $\theta \approx 30^\circ$ estimate the value of R .

Solution: Rest mass of electron is $m_0 c^2 = 0.511 MeV$ which is negligible compare to kinetic energy $K = 500 MeV$

So momentum $pc = E = K \Rightarrow p = \frac{K}{c} = \frac{500 MeV}{3 \times 10^8 m / sec} \times \frac{1 Joule}{6.2 \times 10^{12} MeV} = 2.7 \times 10^{-19} kg - m / sec$

The de Broglie relation gives $\lambda = \frac{h}{p} = \frac{6.6 \times 10^{-34} Joule - sec}{2.7 \times 10^{-19} kg - m / sec} = 2.4 \times 10^{-15} m$

$$R = \frac{\lambda}{\theta} = \frac{2.4 \times 10^{-15}}{0.53 rad} = 4.5 \times 10^{-15} m = 4.5 \text{ Fermi}$$

Uncertainty principle and radius of nucleus.

Using Heisenberg uncertainty principle, it can be shown why electrons can not be within nucleus.

$\Delta x \Delta p_x \approx \hbar \Rightarrow \Delta p_x = \frac{\hbar}{\Delta x}$ the uncertainty in position can be equal to radius of nucleus so

$$\Delta x \approx R = 10^{-14} \Delta p_x = \frac{6.6 \times 10^{-34}}{10^{-14}} = 6.6 \times 10^{-20} \Rightarrow p_x = 6.6 \times 10^{-20} kg - m / sec$$

$$E = \sqrt{p_x^2 c^2 + m_0^2 c^4} \Rightarrow E = \sqrt{(6.6 \times 10^{-20} \times 3 \times 10^8 \times 6.24 \times 10^{12})^2 + (0.511)^2} \text{ MeV}$$

$\sqrt{(124)^2 + (0.511)^2} = 124 \text{ MeV}$ which is very large energy to achieve this energy electron need to have speed more than light so it cannot exist within nucleus

1.2 Fermi Model

A multi fermion system at zero kelvin temperature

Fermions cannot occupy the same state, even at zero temperature. At zero temperature the first fermion sets in the lowest state in the potential well, each next fermion sets in the lowest free state in the well.

Consequently, n fermions occupy the n lowest-energy states in the potential well. The energy of the highest occupied state is called the Fermi energy.

At zero temperature all states up to the Fermi energy are occupied, while all states above the Fermi energy are free. The total energy of the system is equal to the sum of energies of the occupied states.

Assumption of Fermi gas model

The Fermi gas model defines properties of a system of non-interacting fermions in an infinite potential well.

The model predicts gross properties of various quantum-mechanical systems, for example electrons in metals, or nucleons in nuclei.

The model assumes that all fermions occupy the lowest energy states available to them up to the Fermi energy, and that there is no excitations across the Fermi energy (i.e. zero temperature).

In nuclei the model assumes that protons and neutrons are independent fermion filling two separate potential wells. The model assumes, however, common Fermi energy for the protons and neutrons in stable nuclei.

If Fermi energy for protons and neutrons are different then the β decay transforms one type of nucleons into the other until the common Fermi energy (stability) is reached.

Mathematical discussion of model

If a particle of mass m is confined in infinite box with parameter (L_x, L_y, L_z) then energy is

$$\text{given by } E_{n_x, n_y, n_z} = \frac{\pi^2 \hbar^2}{2m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right) \text{ where } n_x = 1, 2, 3 \dots n_y = 1, 2, 3 \dots n_z = 1, 2, 3 \dots$$

for simplicity the $L_x = L_y = L_z = L$ $E_{n_x, n_y, n_z} = \frac{\pi^2 \hbar^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2)$,

where the momentum is given $p_x = \frac{n_x \pi \hbar}{L}$ $p_y = \frac{n_y \pi \hbar}{L}$ $p_z = \frac{n_z \pi \hbar}{L}$

let us consider E_F is fermi energy $E_F = \frac{p_F^2}{2m} \Rightarrow p_F = \sqrt{2mE_F}$

$$p_x^2 + p_y^2 + p_z^2 \leq p_F^2 \Rightarrow \frac{\pi^2 \hbar^2}{L^2} (n_x^2 + n_y^2 + n_z^2) \leq p_F^2 \Rightarrow (n_x^2 + n_y^2 + n_z^2) \leq \frac{p_F^2 L^2}{\pi^2 \hbar^2} = R^2$$

where $R = \frac{p_F L}{\pi \hbar}$ assume to be radius of sphere if (n_x, n_y, n_z) are representing coordinates along

x, y and z The volume of the sphere is the number of states up to the Fermi

momentum since there is one state per unit volume.

To count number of state (n_x, n_y, n_z) we use a trick First, remember that $n_x > 0$ and $n_y > 0$ and $n_z > 0$, thus we have to take only $1/8$ of the sphere cut by positive x, y and z axes.

Next, nucleons have spins. Thus we have two states per unit volume as defined by two orientation of spins.

With these corrections we can calculate the number of occupied states

$$n = 2 \frac{1}{8} \frac{4}{3} \pi R^3 = \frac{1}{3} \pi \left(\frac{p_F L}{\pi \hbar} \right)^3$$

Calculation of Fermi Momentum

$$p_F = \hbar (3\pi^2)^{1/3} \frac{n^{1/3}}{L} = \hbar (3\pi^2)^{1/3} \left(\frac{n}{L^3} \right)^{1/3} = \hbar (3\pi^2)^{1/3} \left(\frac{n}{V} \right)^{1/3}$$

$$\text{If } R = R_0 A^{1/3} \quad V = \frac{4}{3} \pi (R_0 A^{1/3})^3 = \frac{4}{3} \pi R_0^3 A \quad \text{if there is } Z \text{ number of protons}$$

$$\text{Fermi momentum for protons } p_F = \hbar (3\pi^2)^{1/3} \left(\frac{n}{V} \right)^{1/3} = \frac{\hbar}{R_0} \left(\frac{3^2 \pi}{4} \right)^{1/3} \left(\frac{Z}{A} \right)^{1/3}$$

$$\text{Fermi momentum for neutrons } p_F = \hbar (3\pi^2)^{1/3} \left(\frac{n}{V} \right)^{1/3} = \frac{\hbar}{R_0} \left(\frac{3^2 \pi}{4} \right)^{1/3} \left(\frac{A-Z}{A} \right)^{1/3}$$

Where Z is atomic number and A is mass number

Assuming $Z/A \sim 1/2$ and $R_0 = 1.2 \text{ fm}$ the estimate for the Fermi momentum is

$$p_F = \frac{\hbar}{R_0} \left(\frac{3^2 \pi}{4} \right)^{1/3} \left(\frac{Z}{A} \right)^{1/3} \approx 250 [MeV/c] \text{ and speed } \beta_f = \frac{v_f}{c} \approx \frac{p_F c}{m_p c^2} \approx \frac{250}{940} = 0.27$$

Estimation of Fermi Energy

Fermi energy can be estimated from the Fermi momentum.

Fermi energy depends on the density of states and is different for protons and for neutrons since in general there is a different number of particles in the proton and neutron well.

The Fermi energy for protons is
$$E_F = \frac{p_F^2}{2m_p} = \frac{(\hbar c)^2}{2R_0^2 m_p c^2} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} \left(\frac{Z}{A} \right)^{\frac{2}{3}}$$

The Fermi energy for neutron is
$$E_f = \frac{p_F^2}{2m_p} = \frac{(\hbar c)^2}{2R_0^2 m_p c^2} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} \left(\frac{A-Z}{A} \right)^{\frac{2}{3}}$$

For the case of even number of protons and neutrons $Z/A \approx (A-Z)/A \approx 1/2$ and $R_0 = 1.2 fm$ the Fermi energy is $E_f \sim 33 MeV$.

Spin and Magnetic Moment

Proton and neutrons, like electrons, are fermions with spin quantum numbers of $s = \frac{1}{2}$. This

means they have spin angular momenta $\vec{S}$ of magnitude $|\vec{S}| = \sqrt{s(s+1)}\hbar = \sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right)}\hbar = \frac{\sqrt{3}}{2}\hbar$

and spin magnetic quantum number of $m_s = \pm \frac{1}{2}$.

As in case of electrons, magnetic moments of nucleus are associated with the spins of protons and neutrons. In nuclear physics magnetic moments are expressed in **nuclear magnetons** (μ_N),

where nuclear magneton $\mu_N = \frac{e\hbar}{2m_p} = 5.051 \times 10^{-27} J/T = 3.152 \times 10^{-8} eV/T$ and m_p is the proton mass.

In atomic physics we have defined **Bohr magnetons** $\mu_B = \frac{e\hbar}{2m_e}$ where m_e is the electron mass.

The nuclear magneton is smaller than the Bohr magneton by the ratio of the proton mass to the electron mass which is 1836.

$$\frac{\mu_B}{\mu_N} = \frac{m_p}{m_e} = 1836.$$

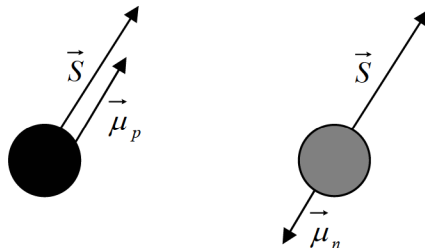
The spin magnetic moments of the proton and neutron have components in any direction of

Proton $\mu_{pz} = +2.793\mu_N$

Neutron $\mu_{nz} = \mp 1.913\mu_N$

There are two possibilities for the signs of μ_{pz} and μ_{nz} , depending on whether m_s is $-\frac{1}{2}$ or $+\frac{1}{2}$.

The $\pm$ sign is used for μ_{pz} because $\vec{\mu}_{pz}$ is in the same direction as the spin $\vec{S}$, whereas $\mp$ is used for μ_{nz} because $\vec{\mu}_{nz}$ is opposite to spin $\vec{S}$.



Note: For neutron, magnetic moment is expected to be zero as $e = 0$, but $\mu_n = \mp 1.913\mu_N$.

At first glance it seems odd that the neutron, with no net charge, has spin magnetic moment. But if we assume that the neutron contains equal amounts of positive and negative charge, a spin magnetic moment arises if these charges are not uniformly distributed. The magnetic moment of neutron further explained by quark model.

Two type of quark are necessary to build proton and neutrons, there are up quark with charge of $\frac{2}{3}e$ and the down quark with $-\frac{1}{3}e$. The proton consists of two up quark and one down quark so that total charge $+e$ and neutron on other hand is built up of one up quark and 2 down quark so that its total charge is zero but there is magnetic dipole moment due to magnetic moment of individual quark.

Thus, we can say that neutron has physical significance of negative charges because magnetic moment is opposite to that of its intrinsic spin angular momentum.

Angular Momentum of Nucleus

The hydrogen nucleus ^1H consists of a single proton and its total angular momentum is given by $|\vec{S}| = \frac{\sqrt{3}}{2}\hbar$. A nucleon in a more complex nucleus may have orbital angular momentum due to motion inside the nucleus as well as spin angular momentum. The total angular momentum of such a nucleus is the vector sum of the spin and orbital angular momenta of its nucleons, as in the analogous case of the electrons of an atom.